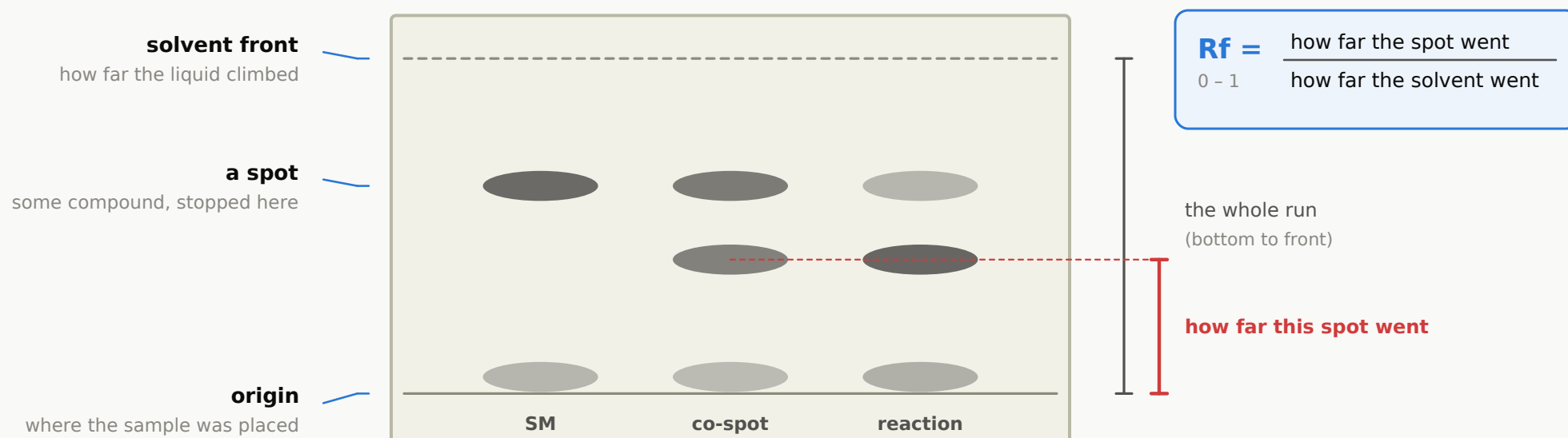


Reading a TLC plate

Seventeen questions, answered with evidence rather than opinion

Prepared by Tejas Ghatule · every source is linked on the last two pages



THE QUESTIONS, IN THE ORDER A REASONABLE PERSON WOULD ASK THEM

- | | | | |
|---|-----|---|-----|
| 1 What am I actually looking at? | p2 | 2 What could a machine tell me that I can't see? | p3 |
| 3 Hasn't somebody already built this? | p4 | 4 Show me how they actually work | p7 |
| 5 And what does the software do with the photo? | p8 | 6 How accurate are they, really? | p9 |
| 7 Then why is all this free software unused? | p10 | 8 What does a chemist know that a camera doesn't? | p11 |
| 9 Where would that missing knowledge come from? | p12 | 10 Can't I just show the plate to ChatGPT? | p13 |
| 11 So why hasn't AI cracked this already? | p14 | 12 What if we're wrong about all this? | p15 |
| 13 Who would actually pay for it? | p16 | 14 Is there a buyer that didn't exist before? | p17 |
| 15 So what exactly would we build? | p18 | 16 What is it worth, honestly? | p19 |
| 17 How would we know if it worked? | p20 | | |

AND THE THREE ANSWERS, IN ONE LINE EACH

Measuring is done, and free

More than thirty tools already turn a plate photograph into spot positions and intensities. Several are free, one drives a purification system, and almost nobody uses any of them.

Naming is empty

No product reads a plate in the context of the reaction it came from and says which spot is the product. That is the gap, and it is a narrow one.

The buyer is changing

A chemist reads a plate in five seconds and never needed help. A robot cannot read one at all, and about four billion dollars has just gone into robots that run reactions.

TWO CORRECTIONS TO THE EARLIER VERSIONS OF THIS REPORT

The first version said no published paper trains a neural network on TLC plate images. The second version corrected that to one. Both were wrong: there are at least four, the earliest from 2018, and one of them is by the same group that wrote the best-known free TLC software. The first version also described this as an almost empty field. Looking harder turned up more than thirty distinct tools that read a plate from an image, going back to 2007. The field is small — but it is not empty, and it was not waiting for us.

Q1 What am I actually looking at?

A sheet of silica, a line where the sample started, and a line where the solvent stopped

Everything in this report is about turning that picture into something a computer can act on. It is worth being exact about what is in the picture and what is not, because almost every disagreement later traces back to this page.

A spot is a compound that stopped

The solvent climbs the plate and drags the sample with it. Anything that sticks harder to the silica lags behind; anything that prefers the solvent runs ahead. A spot is simply where one compound stopped.

R_f is a ratio, not a property

How far the spot went, divided by how far the solvent went. It is between 0 and 1, it has no units, and it depends on the solvent, the plate, the temperature and the humidity — so the same compound gives a different R_f in a different lab. Page 9 has the numbers.

Darkness stands in for quantity

More material absorbs more light, so a darker spot means more of something. The relationship is a known optical law, which is why a machine can measure it honestly. It is not linear in the raw pixel value, which is why naive brightness measurements go wrong.

The plate does not say what anything is

There is no label attached to a spot. Which lane held the starting material, what reaction was being run, what was expected — none of that is in the image. This single fact drives most of this report.

AND HERE IS WHY IT IS HARDER THAN IT LOOKS

Half the compounds are invisible

Under a UV lamp only compounds that absorb at that wavelength show up. A standard teaching reference puts it plainly: most functional groups do not absorb and will not appear, even though they are still there. Iodine staining catches roughly half of what you might encounter, and the stain fades — which is why chemists trace spots in pencil before the evidence disappears.

One spot does not mean one compound

Two compounds with the same R_f sit on top of each other. Regioisomers and diastereomers routinely do. On a public forum a plate argued over as two or three spots turned out, by GC-MS, to be four compounds: 57.6% product, 36.6% alcohol, 3.6% ether, 1.1% aldehyde.

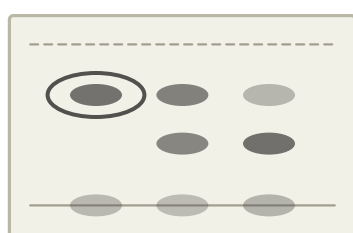
Experienced chemists disagree about real plates

A chemist posts a plate showing a green fluorescent centre with a blue edge and asks: one compound or two? The experts split. One says a single component; another says run it on HPLC with a diode-array detector and check peak purity. Days later the poster changes solvent and finds two.

So the honest starting position is that a TLC plate is a fast, cheap, deliberately approximate measurement. Nobody is trying to build an instrument that beats HPLC. The question is whether the approximate answer a chemist already reads by eye can be captured, recorded and reasoned about by a machine.

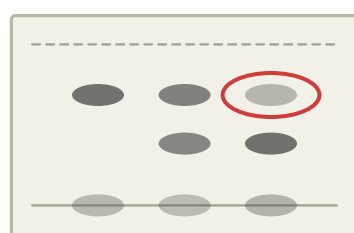
AND THIS IS WHAT A CHEMIST ACTUALLY DOES WITH IT

1 look at the reference lane



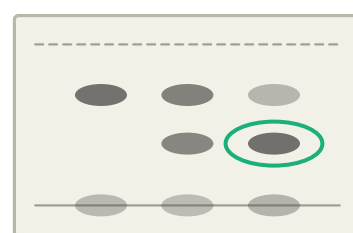
this is what I started with

2 has it gone?



faint at the same height — nearly consumed

3 is there something new?



strong, and matched in the co-spot lane

4 verdict

- starting material: nearly gone
- product: formed, now major
- nothing unexplained

give it one more hour

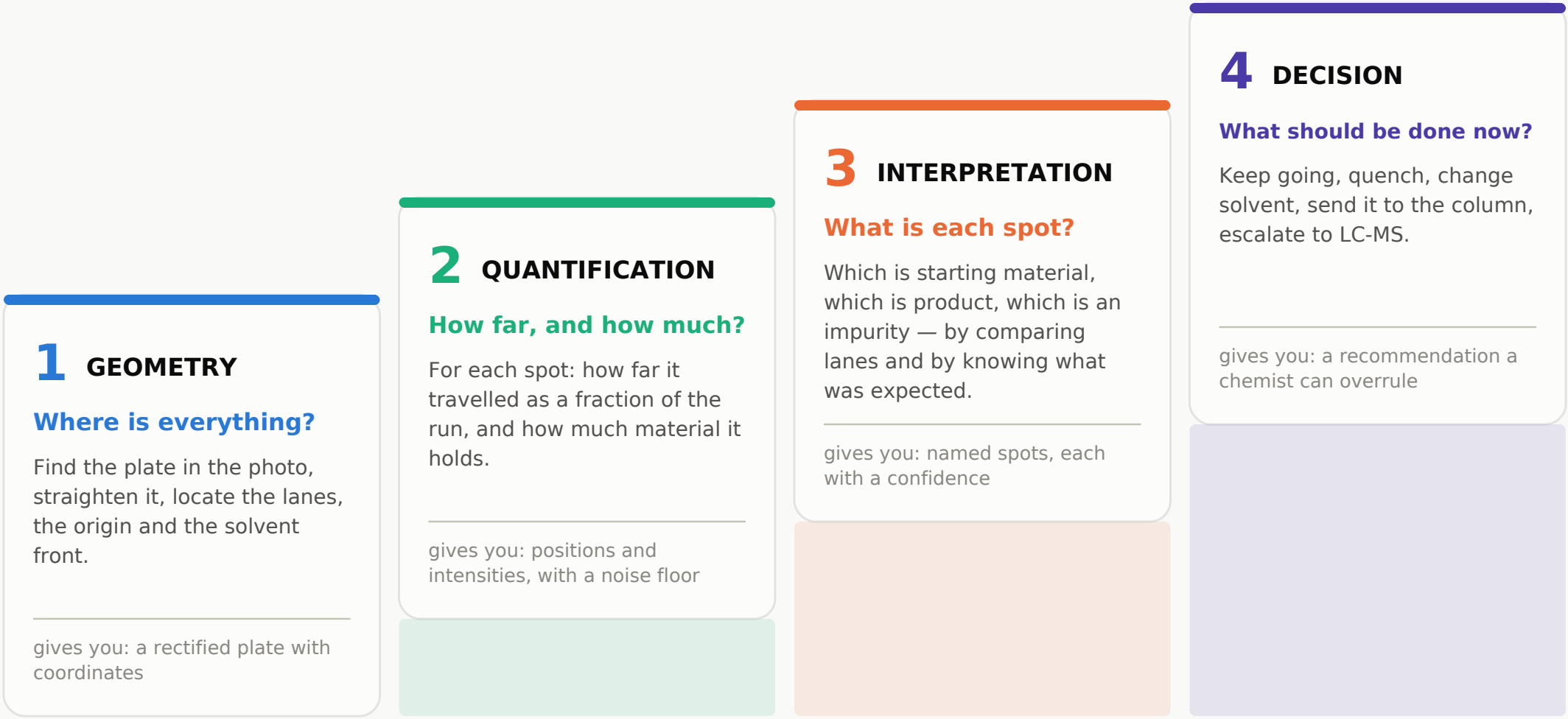
elapsed time: about five seconds

Four comparisons and a verdict, in about five seconds, for free. Every page that follows is really about one question: which part of that is worth handing to a machine, and why nothing built so far has managed it.

Q2 What could a machine tell me that I can't see?

Nothing at the bottom of the ladder. Possibly quite a lot at the top.

Reading a plate is not one job, it is four stacked on each other. Each needs a different kind of evidence and fails in a different way — and every player in this field stopped at a different rung.



MEASUREMENT — the picture answers this

JUDGEMENT — the picture alone cannot

A chemist does all four in about five seconds, and does levels 3 and 4 far better than any software. So the value of automating levels 1 and 2 alone is close to zero for a person standing at a bench — which is exactly what the next few pages show happened.

WHERE EACH APPROACH IN THIS REPORT STOPPED

who	1	2	3	4	what they actually delivered
A chemist's eye	●	◐	●	●	in about five seconds, for free
Scanning densitometers	●	●	○	○	levels 1-2 removed by hardware
Imaging cabinets	●	●	○	○	the same, from a photograph, in a box
Phone apps and web tools	◐	◐	○	○	spot finding and Rf from an ordinary photo
ImageJ, by hand	◐	◐	○	○	the same, with the human doing the geometry
Interchim app + puriFlash	◐	◐	○	◐	spots from a photo, straight into a purification method
visionCATS AI Tools (2026)	●	●	◐	○	classifies a whole sample against trained classes
GAN + CNN species ID (2024)	○	○	◐	○	names the plant, skipping levels 1 and 2 entirely
GPT-4 eluent agent (2024)	○	○	○	◐	picks the next solvent from Rf values typed in as text
What we are building	●	●	●	◐	measured 1-2, named 3, proposed 4

● does this properly ◐ does part of it ○ does not attempt it

Q3 Hasn't somebody already built this?

Yes. More than thirty times, since 2007, and most of it is free.

This is the finding that changed the report. The first version of this research described an almost empty field. Searching properly — GitHub, app stores, Russian and French vendors, gel-analysis software, conference proceedings, university web apps — turned up a crowded, fragmented and largely abandoned space.

2007

the year a high-school science-fair project first turned a TLC photograph into a chromatogram, and published it in a teaching journal

30+

distinct tools that read a plate from an image, found in this survey

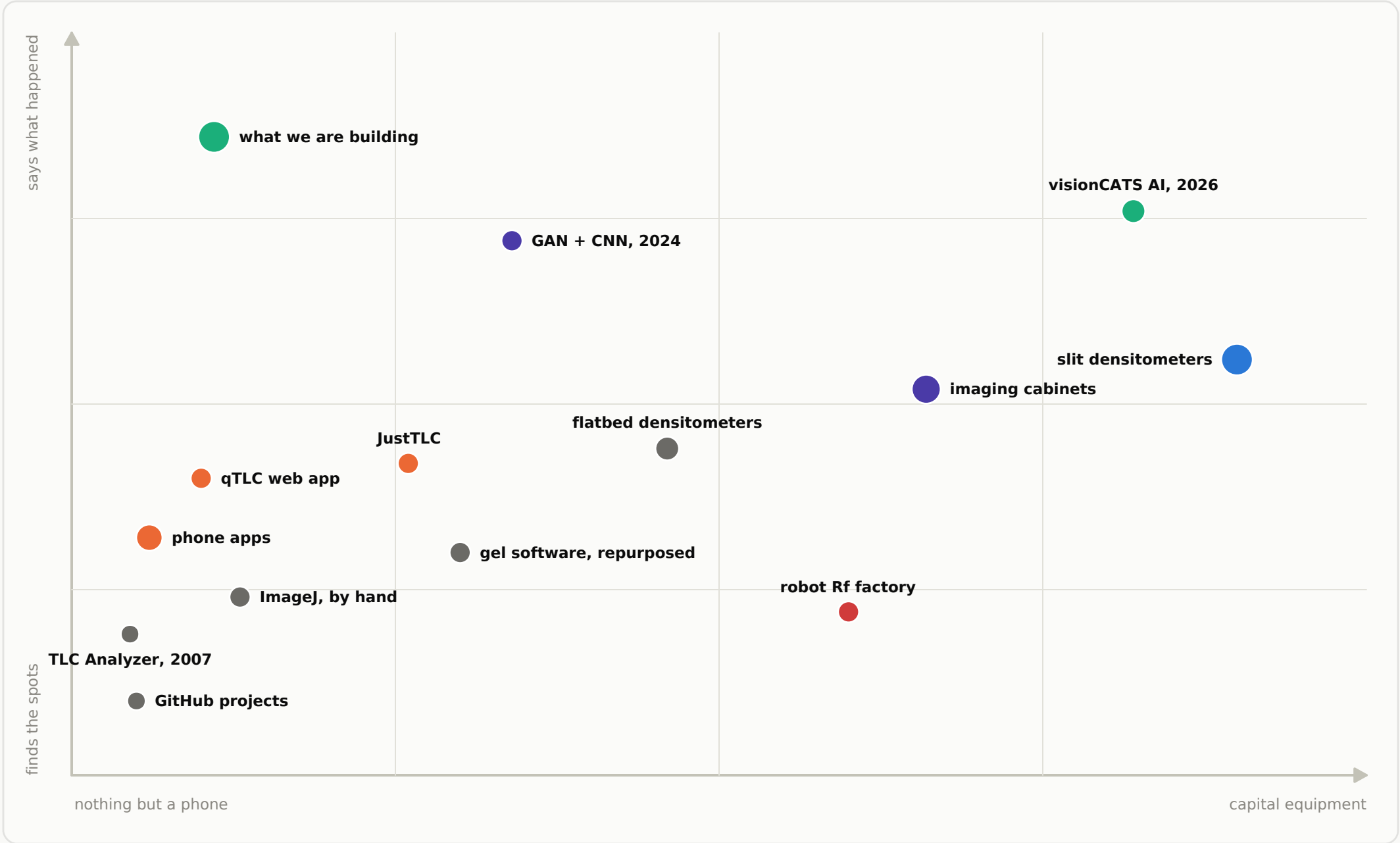
~10

articles a 2023 systematic review could find comparing photo-based reading against a real densitometer

0

of them that name what a spot is

THE WHOLE SPACE ON ONE PICTURE



Read the empty top-left corner. Everything that reads a plate cheaply sits along the bottom — it finds spots and stops. Everything that climbs higher does it by buying hardware, or by skipping the measurement entirely and classifying the whole plate. Nothing occupies the corner where a cheap photograph produces an interpretation, and that is the only place worth us being.

The bottom row is crowded and free

Ten or more tools sit along the bottom edge. They are cheap, they work, and they all stop at the same place.

The right-hand side costs money

Everything that measures well does it by owning hardware — a cabinet, a scanner, a densitometer.

The top-left corner is empty

A cheap photograph producing an interpretation. That is the only square nobody occupies, and the only one worth us entering.

Everything that reads a plate from a photograph

Ten of them. Free unless marked. Coloured bar = still alive.

tool	who / when	price	what it does
TLC Analyzer	Amber Hess · 2007	free	Photo or scan → pick one row of pixels → slide an averaging window along it → four optical-density curves (red, green, blue, brightness) → Excel. The ancestor of everything below. A science-fair project, published in a teaching journal, still downloadable, never updated.
qTLC	University College London · 2018	free	Phone photo (a black box with a 254 nm lamp for fluorescence) → web page → pick lanes and bands → set a minimum peak prominence → two-dimensional integration → calibration line. Reported 2.6–3.8% error on their own test. Web app plus a downloadable version, all components released.
ImageJ / Fiji “Gel Analyzer”	NIH · continuously	free	Draw a rectangle per lane by hand → one density profile per lane → draw the baseline by hand → click inside each peak to integrate. No TLC plugin has ever existed. The community simply treats plate lanes as gel lanes. This is the most-used method in the literature and has no product behind it.
ImageJ + 3D-printed lightbox	FHI 360 · 2024	free	A printed box fixes phone, plate and UV lamp in place → a macro finds each spot automatically → integrated density → regression in Python. Built for screening substandard medicines. Better than 0.99 correlation, 5–6% accuracy. Deliberately applies no background correction — they found it made results worse.
TLCyzer	University of Tübingen · 2022	free	Android app plus a build-it-yourself UV box costing \$36–78 → percentage of declared drug content against a reference standard on the same plate. About 2.8% repeatability in a laboratory. Field-tested in Nigeria — see page 15, where it did not go well.
T.L.C. to Flash & Prep	Advion Interchim · 2024	free	Photograph the plate → automatic spot detection → R _f and a separation-difficulty verdict → transmits a purification method to a puriFlash system over Bluetooth or WiFi. The closest thing in existence to the product one would naively propose, and it is free. About a thousand installs; last updated October 2024.
TLC FlashReader	Advion Interchim	not published	The bench hardware behind that app: a 17×22×32 cm box with three light sources and a camera; auto-trims and auto-rotates the plate; feeds their flash systems and an electronic notebook. Shows the direction the flash-chromatography vendors are taking: the plate as an input to purification, not as a record.
JustTLC / JustQuantify	Sweday, Sweden	\$465 / free	Windows software or a browser page; accepts any camera or scanner image; automatic spot detection, quantification, comparison across images. Cited in over 150 papers, and rated 3.9 from 34 votes with not one written comment.
Chromato	independent · India	free	Android plus OpenCV: capture or upload → crop and rotate → pick lanes and spots → adjust peak prominence → a table of integrated spots. Built for forensic work and presented at a hackathon final. Shipped as an APK on GitHub; never reached an app store.
T.L.C Vision	independent, Spain	free	Photograph a plate, get R _f . Around a hundred installs. Updated July 2025.

Not one of these asks what any spot is. Every single one delivers a position and an amount, and then hands the plate back to the chemist to interpret. That is not a criticism of the software — it is the shape of the whole field, and the reason page 10 exists.

And the rest of it: scanners, gel software, code

The part that never appears in an English-language literature search

tool	who / when	price	what it does
Sorbfil TLC View	Russia · still updated	shareware	The plate is laid face-down on the glass of an ordinary consumer flatbed scanner. Tracks are found, intensity is measured against local background, spot areas are evaluated, a densitogram is produced. Sold in three variants — scanner-based, lighting-chamber and visualizer. Entirely invisible to the English-language literature.
AR2i ChromImage	France · still sold	not published	A flatbed-scanner densitometer for TLC and HPTLC — genuinely image-based rather than a slit scanner. Benchmarked against CAMAG scanners in a 2014 comparison, alongside ImageJ.
TotalLab Phoretix 1D	United Kingdom · active	on request	Gel-analysis software with a published, step-by-step TLC procedure: image adjust, lane creation, background removal, Rf calibration, spot definition, quantification, report. A gel package that quietly documents a full TLC workflow — a good example of how scattered this field is.
UN-SCAN-IT gel	Silk Scientific, USA	on request	Lists TLC among supported sample types; converts any scan or photo into density profiles and band areas. Trading for around 37 years. No Rf function documented.
GelAnalyzer	Hungary · active	free + paid	Automatic lane and band detection, background correction, calibration. Does not claim TLC support; used for it anyway.
LabImage TLC	Kapelan, Germany · legacy	on request	Built to order in 2011 for one company: run distance, sample quantity, macros, reports, 21 CFR Part 11. No longer actively marketed — a reminder that this niche has already produced and buried commercial products.
qtlc (R package)	Belgrade · published 2019	free	True two-dimensional densitometry from archived TIFF images: 21 functions for background fitting, noise, Rf, 2D and 3D plots. On the main R archive, dormant. Not to be confused with the UCL qTLC web app, or with quanTLC.
rTLC / quanTLC	Fichou & Morlock, Giessen	free	Open-source image evaluation and multivariate analysis of planar chromatograms, published in Analytical Chemistry. The most academically serious free option, and the same group later put a neural network on plate images.
appaloosa · pyTLC · TLC_profile	university labs on GitHub	free	Small Python projects: lane profiles to CSV, plate-image processing utilities, an analysis script with a manual. Between zero and seven stars each, a handful of commits, all dormant. This is what the open-source layer of this field actually looks like.
2LabsToGo-Eco / Office Chromatography	Morlock group, Giessen	free designs	Open hardware: a complete miniature planar-chromatography lab built from office peripherals and 3D-printed parts, for about €98. The hardware floor has collapsed. What remains scarce is interpretation.

Two things stand out. Half of this is gel-electrophoresis software being used for a job it does not advertise, which means the demand was real enough for people to improvise. And a whole commercial branch — Russian flatbed-scanner densitometers, a French scanner densitometer, a German package built to order and then abandoned — is invisible from where most of us look.

Q4 Show me how they actually work

Six ways of getting light off a plate and into a number

Everything in the inventory is one of these six, and the differences between them are almost entirely about how much of the variability is removed before the software starts.

1 · A phone, on the bench

nothing is controlled

The cheapest possible capture and the hardest to work with. Room light, angle, distance and white balance all change between one photograph and the next, so nothing can be compared to anything.

2 · A phone and a box

a shoebox, or a 3D-printed one

A shoebox, a cardboard hood or a 3D-printed jig holds the plate, the UV lamp and the phone in fixed positions. TLCyzer, qTLC and the FHI 360 macro all do this. It costs almost nothing and it is what makes the numbers comparable.

3 · A flatbed scanner

plate laid face-down on the glass
the scanner's own lamp is the light source

Put the plate face-down on the glass. The scanner's own lamp and moving sensor give a fixed geometry and even lighting for free. Russian and French vendors sell densitometers built on exactly this.

4 · An imaging cabinet

fixed geometry

the box removes the variability

A camera bolted at a known distance above the plate, with defined 254 nm, 366 nm and white illumination. CAMAG, Merck, Syngene and Analytik Jena all sell a version. The box is not optional.

5 · A slit-scanning densitometer

no camera, no photograph — a physical measurement

No camera and no photograph. A monochromator picks a wavelength, a narrow beam is dragged across the plate, and a photomultiplier measures what comes back. This is the reference method, and it is a capital instrument.

6 · A robot with a fixed camera

fixed camera

a robot spots and moves every plate the same way

4,944 measurements published, no images

Take the human out of the loop entirely: a robot spots, develops and photographs every plate identically. Built not to read messy plates but to manufacture clean ones.

The ordering is not an accident. Read left to right and each rig removes one more source of variability, and costs one step more. The interesting question for us is whether the software can be made good enough that a chemist can stay at step 1 or 2 — because every step beyond that is a purchase, and purchases are where all the free tools stopped being used.

rig	what it costs	what it removes	who uses it
1 phone on the bench	nothing	nothing	anyone in a hurry
2 phone and a box	\$36–78 in parts	lighting, distance and angle	TLCyzer, qTLC, the FHI 360 macro
3 flatbed scanner	a scanner you already own	the same, plus the camera lens itself	Sorbfil, AR2i ChromImage
4 imaging cabinet	capital, price not published	everything except the plate	CAMAG, Merck, Syngene
5 slit densitometer	capital, price not published	the photograph entirely	regulated quality control
6 robot and fixed camera	a robot	the human	one university platform, once

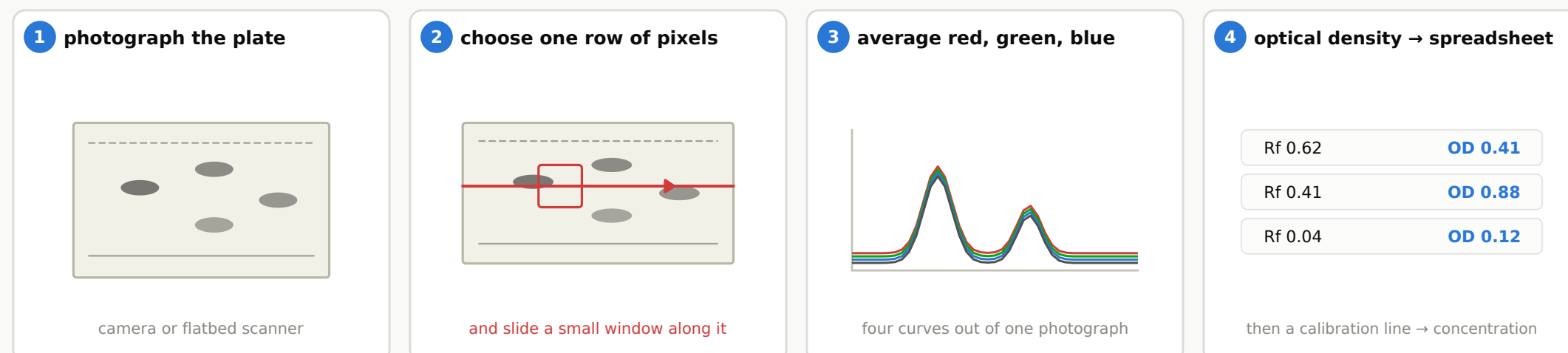
Q5 And what does the software do with the photo?

Almost always the same four steps — and step three is a number the user has to guess

Three worked examples, drawn out. They span eighteen years and three continents, and they are recognisably the same procedure.

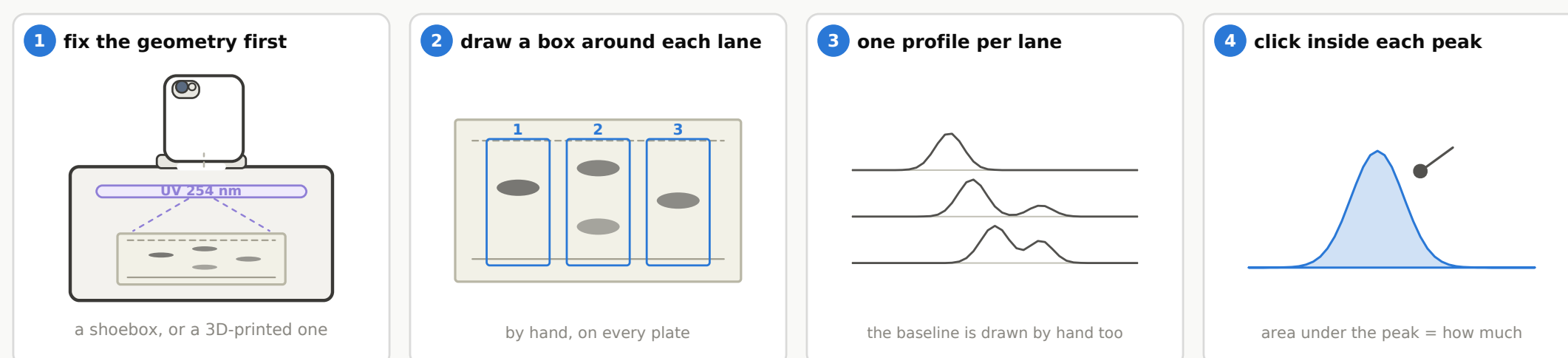
TLC Analyzer, 2007

The original. A photograph, one chosen row of pixels, and a sliding average that imitates what a scanning densitometer does mechanically.



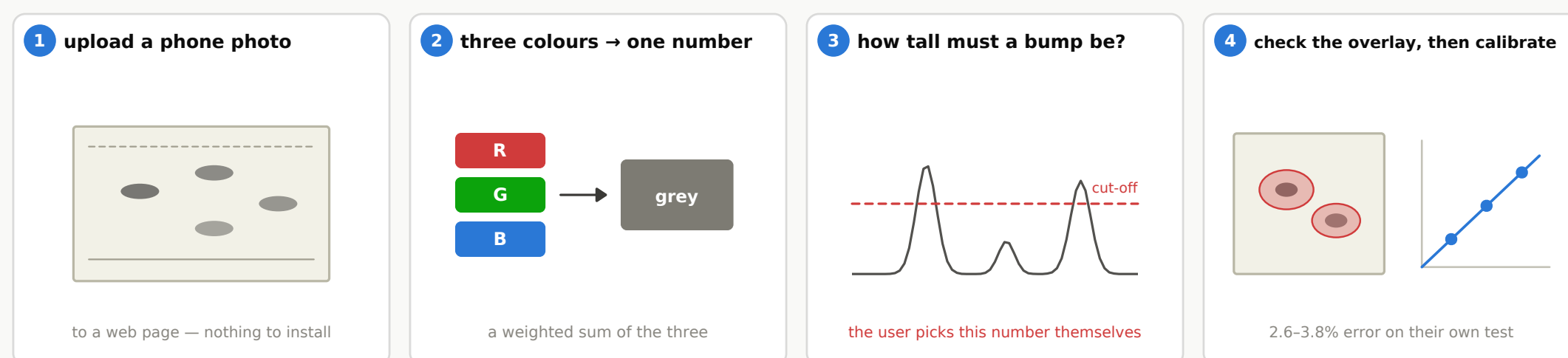
ImageJ, the way most papers actually do it

No TLC plugin exists, so the plate is treated as an electrophoresis gel. Every geometric decision is made by hand, on every plate.



qTLC, University College London, 2018

A web page. Colour is collapsed to one grey number, bumps above a user-chosen height become spots, and the recognised area is painted back so it can be checked.



WHAT EVERY ONE OF THEM HAS IN COMMON

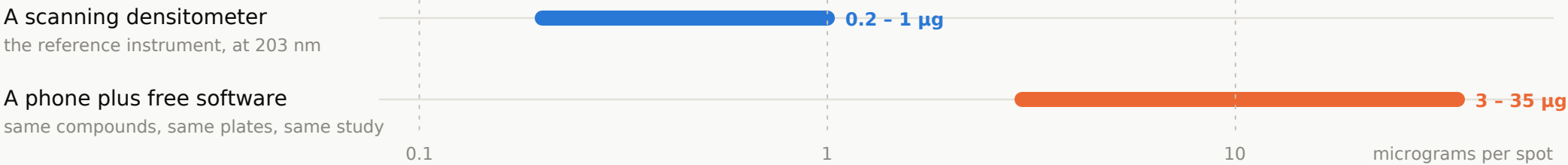
Three things are true of all of them. The geometry is done by the human, not the machine. Somewhere there is a threshold the user has to choose, and choosing it differently gives a different answer. And the output is always a number attached to a position — never a name. Eighteen years, and the shape has not changed.

Q6 How accurate are they, really?

Good enough to be useful. Fifteen to thirty times less sensitive than the instrument they replace.

One 2026 paper ran the same compounds on the same plates through both a scanning densitometer and a smartphone with ImageJ. It is the cleanest comparison available.

THE SAME PLATES, TWO WAYS OF READING THEM



Both methods agreed with each other statistically, both had relative deviations under 2%, and both recovered 98–102% of what was there. The phone was simply blind to anything below about three micrograms per spot, where the densitometer could still see one fifth of a microgram. For screening a reaction that difference rarely matters. For proving a trace impurity is absent, it decides everything.

AND THE NUMBERS EVERYONE HAS TO LIVE WITH

what varies	measured value	source
Rf inside one laboratory	relative deviation under 20%, worse below Rf 0.2	IAEA inter-laboratory study
Rf inside one lab, in the well-behaved band 0.3–0.7	variation under 0.1	IAEA inter-laboratory study
Relative Rf between laboratories	mostly under 11% — but 17%, 19% and 31% for three of them	IAEA inter-laboratory study
hRf under controlled humidity	26 ± 1 (observed range 25–28)	humidity study, 2023
Assay by proper densitometry, two labs	1.90% within a lab, 2.74% between labs	AOAC study
Smartphone quantification, in a laboratory	4.46% intermediate precision	TLCyzer, 2022
The same app in the field, against HPLC	8.4% median, 16.4% mean deviation	Nigeria field trial, 2024
Antibiotics by phone and iodine vapour	0.7–0.8% repeatability, 98–99% recovery	RSC Advances, 2021
Two drugs, phone versus densitometer	no statistically significant difference	BMC Chemistry, 2024

The inter-laboratory study's conclusion is the one that matters commercially: individual Rf values vary substantially between laboratories, and position measured against a reference run on the same plate is far more stable than Rf itself. The pharmacopoeias agree — the official method is a comparison against a standard on the same plate, not a lookup against a published number. Software selling beautifully precise absolute Rf is answering a question the discipline does not ask.

THERE IS ALSO A REAL ALGORITHM LITERATURE NOBODY IN THIS FIELD CITES

A group in Porto spent a decade on the specific problem of finding lanes and bands automatically in TLC images — wavelet smoothing of the intensity profile, then morphological candidate detection, then validation by lane width, spacing and intensity. They tested on 66 images with 651 lanes and 169 images with 1,422 lanes, and beat three earlier methods. Their application was screening for a genetic disease from urine, not chemistry, and the work sits in conference proceedings — which is exactly why nobody reading the smartphone-TLC literature has ever seen it. They also report the failure mode honestly: their method discards the top quarter of every image because biological samples pile up there, and one documented miss was a band inside the discarded strip.

Then why is all this free software unused?

Because it delivers the half a chemist could already do in five seconds

This is the most uncomfortable page in the report, and the one that should be read before any decision.

~1,000
installs of the free Interchim app that photographs a plate, finds the spots and sends a purification method to their flash system over Bluetooth. Last updated October 2024.

~100
installs of an independent app that does automatic Rf from a photograph.

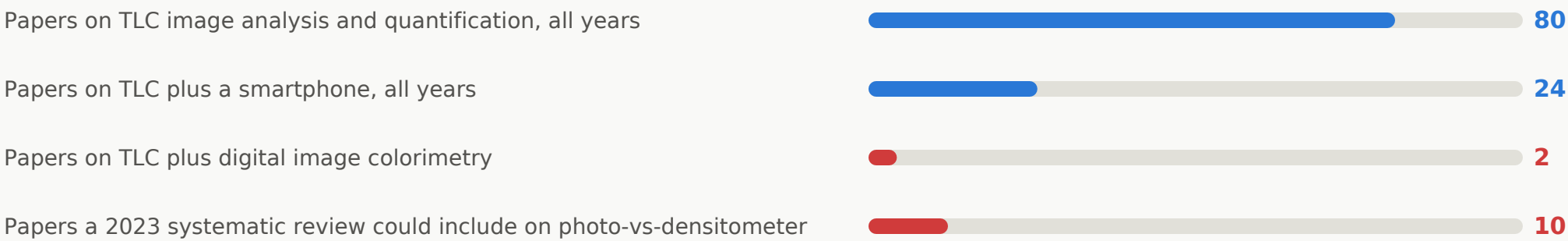
3.9 / 34
the rating of the one paid package, from thirty-four votes — and not one written comment in the whole listing.

3
total ratings on another free plate-reading app, last updated 2015.

THE SINGLE HARDEST FACT IN THIS REPORT

A flash-chromatography company built and gave away the exact product a newcomer would propose. It works. It is free. It connects to hardware they sell, so they had every commercial reason to push it. It has about a thousand installs worldwide, in a technique used in essentially every synthesis laboratory on earth.

AND THE RESEARCH SIDE IS SMALL TOO — HERE IS HOW SMALL



Those searches under-count — analytical-chemistry journals are poorly indexed and Chinese-language work is invisible from here. But the direction is unambiguous. A whole systematic review, four databases, found ten articles comparing a photograph against a densitometer. This is a niche, and it has been a niche for twenty years.

SO WHAT WOULD HAVE TO BE DIFFERENT

Deliver a conclusion, not a number
Rf values are the part the chemist already has. A verdict on the reaction is the part they do not.

Cost the chemist nothing to use
Every tool above asks the user to crop, pick lanes, set a threshold. Each request is a reason to close the app and look at the plate instead.

Be worth keeping even when it is wrong
A stored, searchable, dated record of every plate has value on its own — which is the one thing none of these tools tried to sell.

AND HERE IS WHAT HAPPENED TO EACH OF THEM

tool	last sign of life	what that tells you
TLC Analyzer	version 1.1, 2007, never updated	the idea is eighteen years old
TLC Chemistry Tools	last updated 2015	abandoned by its author
LabImage TLC	built to order in 2011, no longer marketed	a commercial product already died here
Interchim's app	last updated October 2024	still alive, and still barely installed
appaloosa, pyTLC, TLC_profile	a handful of commits each, all dormant	the open-source layer is one person, once
qtlc R package	published 2019, dormant	the same, in a different language

Q8 What does a chemist know that a camera doesn't?

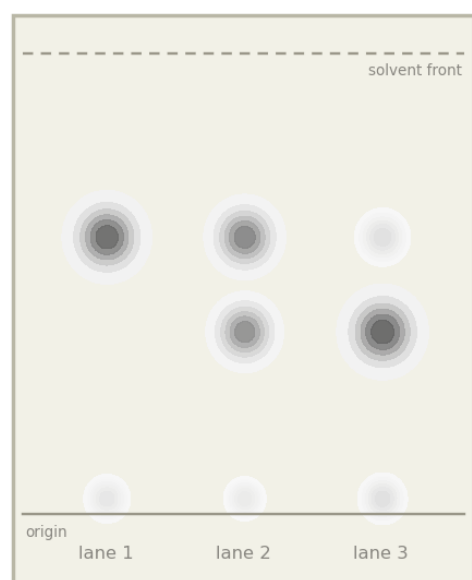
Which lane is which — and without that, the same picture has several correct readings

A spot carries a position and a darkness. It does not carry a name.

THE SAME PIXELS, THREE DIFFERENT ANSWERS

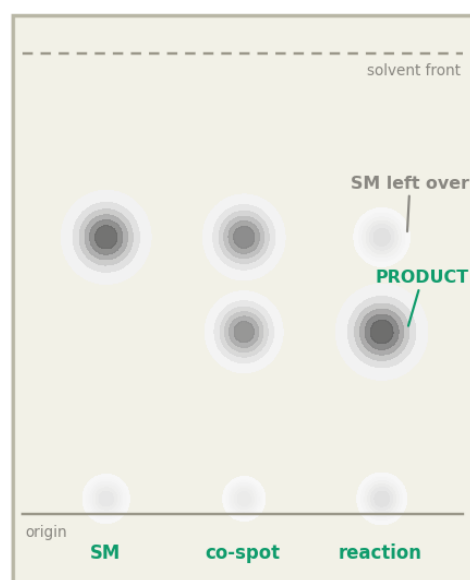
A spot carries a position and a darkness. It does not carry a name. Naming it needs the chemist's context — which is why measuring and interpreting are two separate jobs.

WHAT THE CAMERA RECORDS



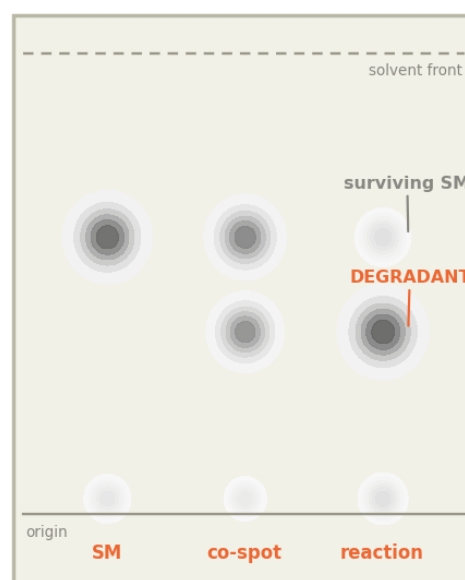
Three lanes. Eight dark regions. Each has a position and a darkness, and that is all the file contains.

READING 1 — it worked



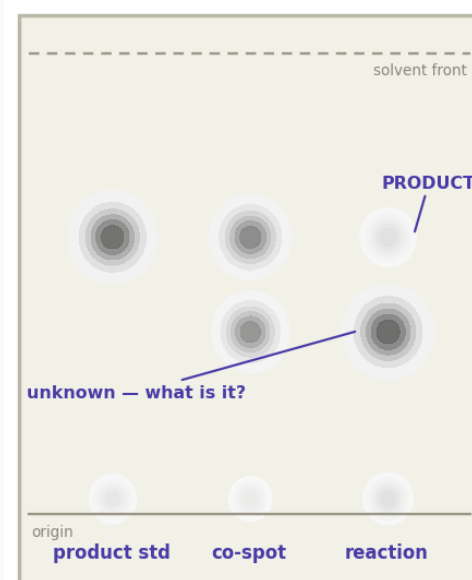
The new spot at 0.41 is the product; the faded 0.62 is unreacted starting material. Nearly done — one more hour.

READING 2 — it went wrong



Identical picture. If that new spot is a break-down product, the reaction failed and the material is being eaten. Stop.

READING 3 — lanes swapped



If lane 1 held the product standard, the 0.62 spot is the product and the big new spot is something nobody asked for.

Everything at levels 1 and 2 is measurement, and measurement is honest work: a spot's position and darkness really are in the file, and careful arithmetic recovers them. Nothing above that is in the file. Which lane held the starting material, what reaction was run, what was expected to form, whether the lanes were spotted left to right or right to left — none of it is visible, and no amount of image processing will find it.

WHICH IS WHY EVERY TOOL ON PAGES 5 AND 6 STOPPED WHERE IT DID

They did not stop because the authors lacked ambition. They stopped because the next step is not an image problem at all. Going further means asking the chemist for something, or reading it off the plate's own handwriting, or inferring it from how the lanes relate to each other. The next page is about those routes.

THREE WAYS THIS GOES WRONG IN A REAL LAB, NOT IN A DIAGRAM

The lanes were spotted right to left

Some chemists load the reference last, on the right. A system that assumes a fixed order will confidently invert every conclusion on the plate and give no sign that it has.

The plate was re-used or re-run

A second development in a different solvent leaves both sets of spots on the plate. The picture now contains two experiments and no marker saying which is which.

The label is wrong

The pencil says “SM” because that is what the chemist meant to spot. Reading the handwriting correctly does not make the handwriting true, which is why every named spot has to carry a confidence and stay easy to overrule.

Two compounds sit on top of each other

Regioisomers and diastereomers routinely share an R_f . The plate shows one spot, the software measures one spot, and both are right. Nothing in the image can reveal the second compound — only a different solvent, or a different instrument, can.

Q9 Where would that missing knowledge come from?

Four places, and none of them is the pixels

The chemist tells us

Lane names, the reaction, what was expected — typed once when the plate is uploaded, or pulled from the notebook entry the plate already belongs to. Cheapest and most reliable, and the source every existing tool skipped.

It is written on the plate

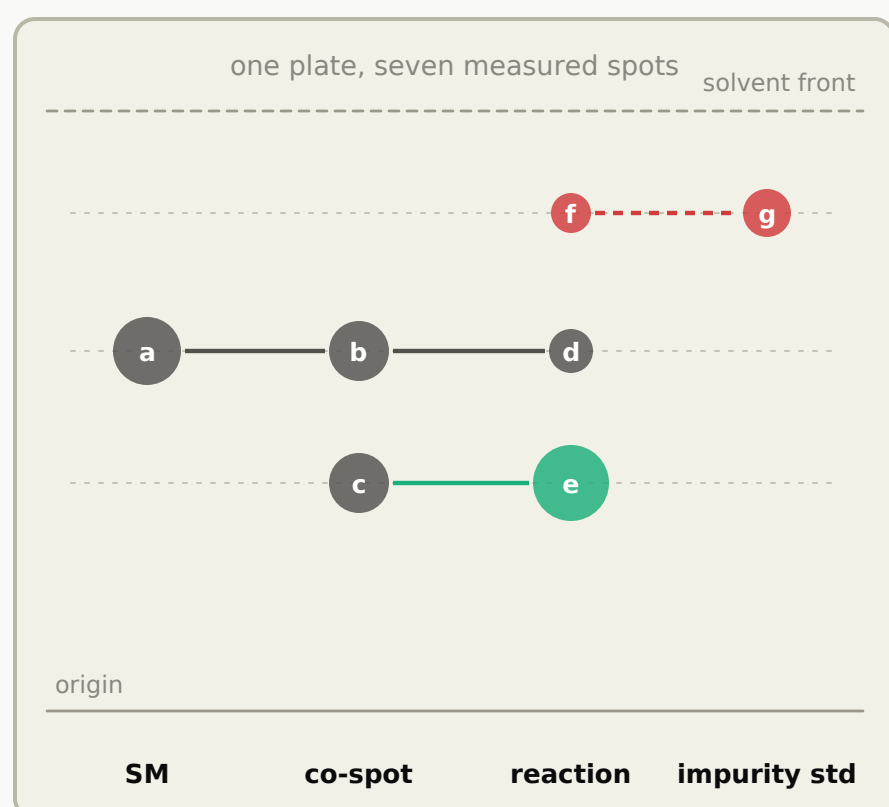
Chemists label lanes in pencil at the top of the plate. That handwriting is readable — but by a language model, not by conventional text recognition. In our own benchmark every text-recognition engine we could run failed on all fourteen crops; a vision-language model read them all.

The other lanes imply it

A spot in the reaction lane that lines up with the starting-material lane is starting material. A spot with no match anywhere is new. This is reasoning about relationships, not about pixels.

Time implies it

Across a series of plates from one reaction, a spot that fades every hour and one that grows every hour identify themselves without anybody naming them.



WHAT THE THIRD ROUTE ACTUALLY RESOLVES

a — b — d line up

The reaction lane still has a spot where the starting material sits, but much fainter. Conclusion: consumed, not finished.

c — e line up

Present in the co-spot lane, strong in the reaction lane, absent from the starting-material lane. Conclusion: this is the product, and it is now the major component.

f — g line up

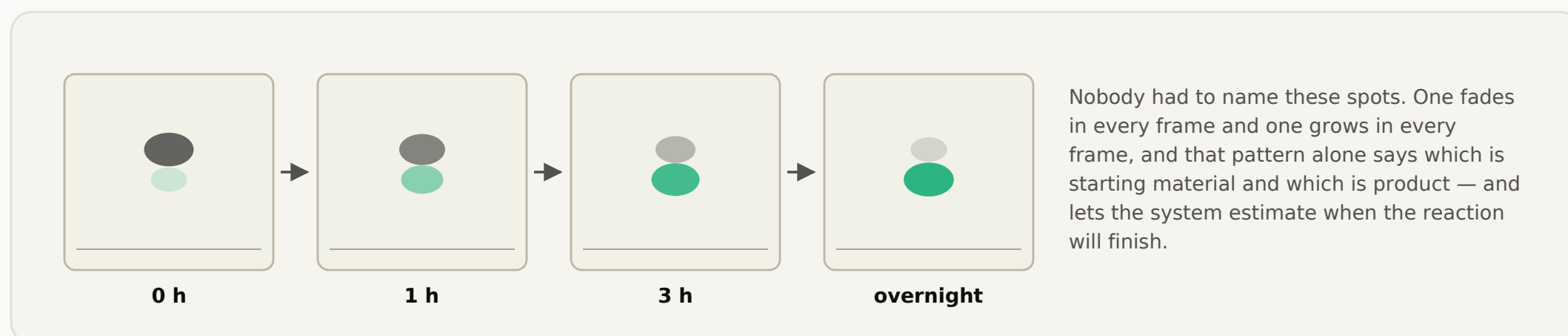
A faint spot matching a deliberately spotted impurity standard. Conclusion: carried in with the batch, not created by the reaction.

nothing else appears

No spot in the reaction lane is unexplained — which is the question a chemist actually cares about.

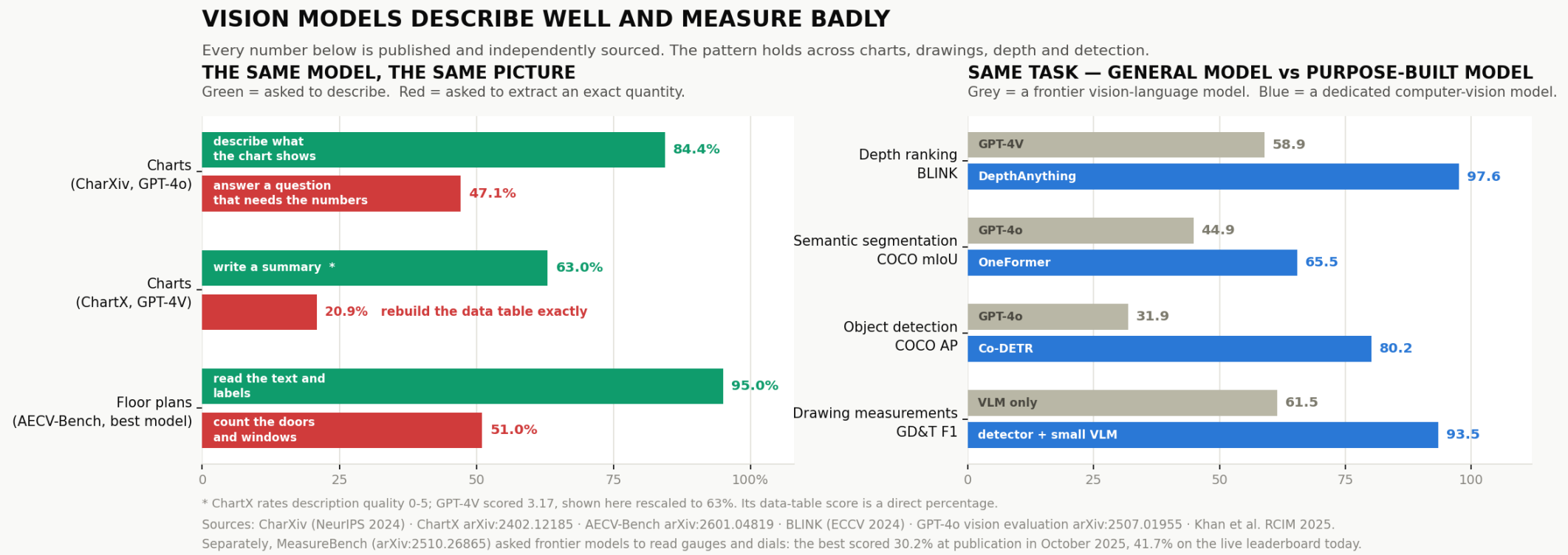
Not one of those four conclusions can be drawn from a single spot. Each comes from a comparison — this lane against that lane, this darkness against that darkness. The measurement layer produces the dots; the conclusions live in the lines between them. A tool that stops at “here are your R_f values” hands over the dots and keeps the part that was worth having.

AND THE FOURTH ROUTE — THE SAME IDEA, STRETCHED ACROSS TIME



For half of it, yes. For the measuring half, it is the worst tool available.

This is not a guess. It is the most heavily benchmarked question in the report.



The pattern holds across every benchmark we could find, and it is not a story about models being weak. It is a story about two different kinds of question. Asked to describe, explain or classify what is in a picture, these models are close to human. Asked to return an exact position, an exact count or an exact value, they degrade sharply — and a small purpose-built model trained on that one task beats them decisively.

AND ON OUR OWN PLATES, THE SAME SPLIT APPEARED

what we asked	classical methods	a vision-language model
Read the handwritten lane labels	0 of 14 crops — every engine we could run	all correct
Find the spots and their positions	6 of 6 verified features on our hardest plate	misses faint spots; positions drift
Measure how much material is in a spot	physical: optical density against a fitted background	no calibrated quantity at all
Say which spot is the product	cannot — there is no rule to apply	correct, given the lane context
Explain the result to a chemist	not applicable	fluent, and accurate when given the numbers

BUT SHOULDN'T WE JUST WAIT FOR THE MODELS TO GET BETTER?

The numbers are moving fast

A benchmark that asks frontier models to read gauges and dials had a best score of 30.2% when it was published in October 2025. The live leaderboard best today is 41.7%. Betting that this stays hard is betting against a visible trend.

But 41.7% is still not a measurement

A reading has to be right nearly every time, not two times in five. And on the same leaderboard one frontier model scores 7.5% — the spread between models is wider than the improvement, which is not what a solved capability looks like.

And measuring is the cheap half

Our measurement code is a few hundred lines of arithmetic with no training and no inference cost. If a model eventually does it better, that component gets swapped and everything above it survives. Page 18 is designed so that this is a swap, not a rewrite.

Four groups tried. There is no training data, and the ones who could have released it published tables instead.

CORRECTING THIS REPORT FOR THE SECOND TIME

Earlier versions said zero, then one. Searching properly turns up at least four pieces of work that put a neural network on planar-chromatography images, the earliest in 2018 — and that one is by the same group that wrote rTLC and quanTLC, the best-known free TLC software. It is still a tiny literature. It is not zero, and each time this report looked harder the number went up.

EVERYTHING WE COULD FIND THAT PUTS A NETWORK ON A PLATE IMAGE

2013 An artificial neural network models HPTLC data for a medicinal plant.

Modelling numbers extracted from plates, not the image itself.

2018 A neural network for planar-chromatography image evaluation — denoising and feature extraction.

Fichou & Morlock, Analytical Chemistry. The first network genuinely applied to plate images, by the authors of rTLC.

2024 A generative network invents synthetic HPTLC plates so a convolutional network can be trained on seven ginger species.

98.7% on the 77 real images. Skips geometry and quantification entirely: one label for a whole plate.

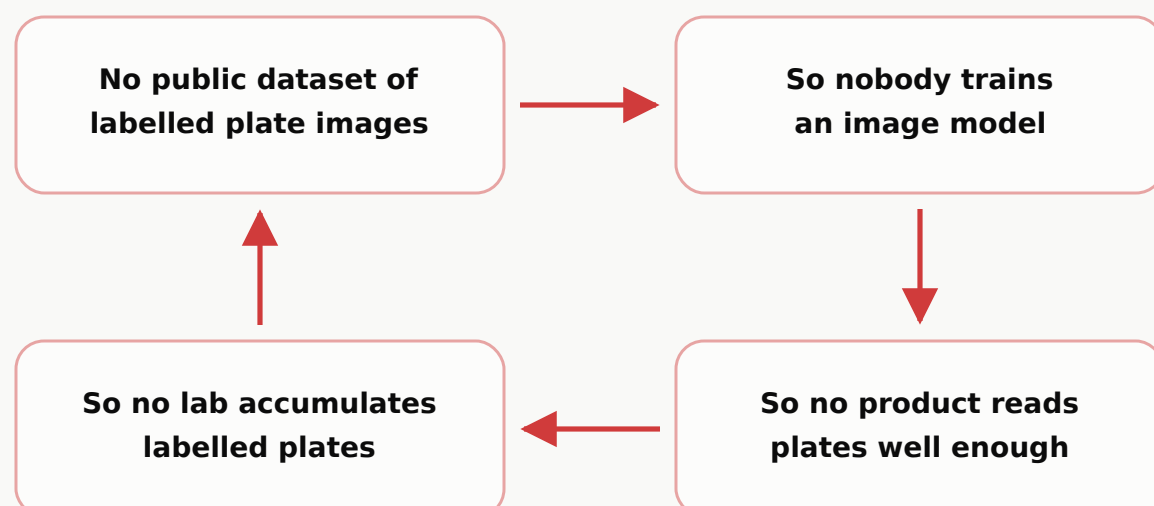
2024 Machine vision quantifies tea saponins from a TLC-based haemolysis assay.

A real machine-vision application, published in a food-science journal where nobody in this field would look.

THE PAPER THAT DOES NOT EXIST

And what is not there is as telling. There is no object-detection paper for TLC spots — no YOLO, no U-Net, no segmentation model, nothing on any preprint server. The standard modern recipe for “find the things in this image” has never been applied to a TLC plate, anywhere, by anyone.

THE REASON IS THE SAME EVERY TIME: THERE IS NO DATASET



We searched every repository that matters. HuggingFace: nothing. Kaggle: nothing — the search collides with car licence plates. Roboflow: nothing. Zenodo and figshare: per-paper supplements only.

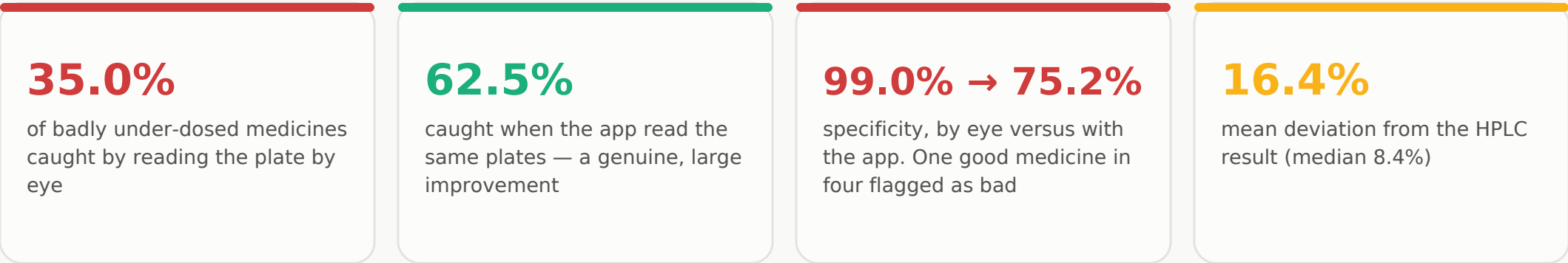
The nearest thing to a corpus is a reference database of HPTLC fingerprint images maintained by the industry association for herbal identity — behind a login, and organised for humans comparing plates by eye, not for training anything. Meanwhile a robot generated 4,944 measurements across 387 compounds and published the table without the photographs.

What if we're wrong about all this?

The one large field trial of automated plate reading failed, and its authors said so

Nigeria, 2024 · Journal of Pharmaceutical Policy and Practice · 260 medicine samples, every result checked against HPLC

TLCyzer is the closest published relative of what we are building: a free smartphone app that photographs a TLC plate and quantifies it against a reference standard. In the laboratory it performed well — about 2.8% repeatability. A German-Nigerian team then put it in the hands of local health workers and pharmacy personnel screening real medicines.



“The TLCyzer app is not fit for deployment in the field at present.” — the authors' own conclusion

WHAT THIS DOES AND DOES NOT MEAN FOR US

- It is real evidence against the whole category**

A carefully built, peer-reviewed, laboratory-validated tool, put in real hands, produced a false-alarm rate its own authors called unacceptable. Anyone claiming this problem is easy has to explain this result first.
- Training moved the numbers a long way**

Deviation from HPLC fell from 22.0% mean before training to 8.3% after. Most of the failure was in how the plate was produced and photographed, not in the arithmetic afterwards. That is a warning about the whole chain, not about the software.
- Their task was harder than ours in one way**

They were asking for an absolute quantity — what percentage of the declared drug is present — from a plate photographed in the field. Page 9 explains why absolute values off a plate are fragile. Our question is comparative: has this spot appeared, has that one gone.
- And easier in another**

Their plates came from a standardised kit, with a reference standard beside every sample and a known list of expected compounds. Ours are ordinary bench plates with handwritten labels. We do not get to claim the easier half of that trade without acknowledging the harder half.
- The honest lesson**

Accuracy in a laboratory tells you very little about accuracy in real use. Any claim we make should be made against plates from our own bench, produced by people who were not trying to make the software look good.

Q13

Who would actually pay for it?

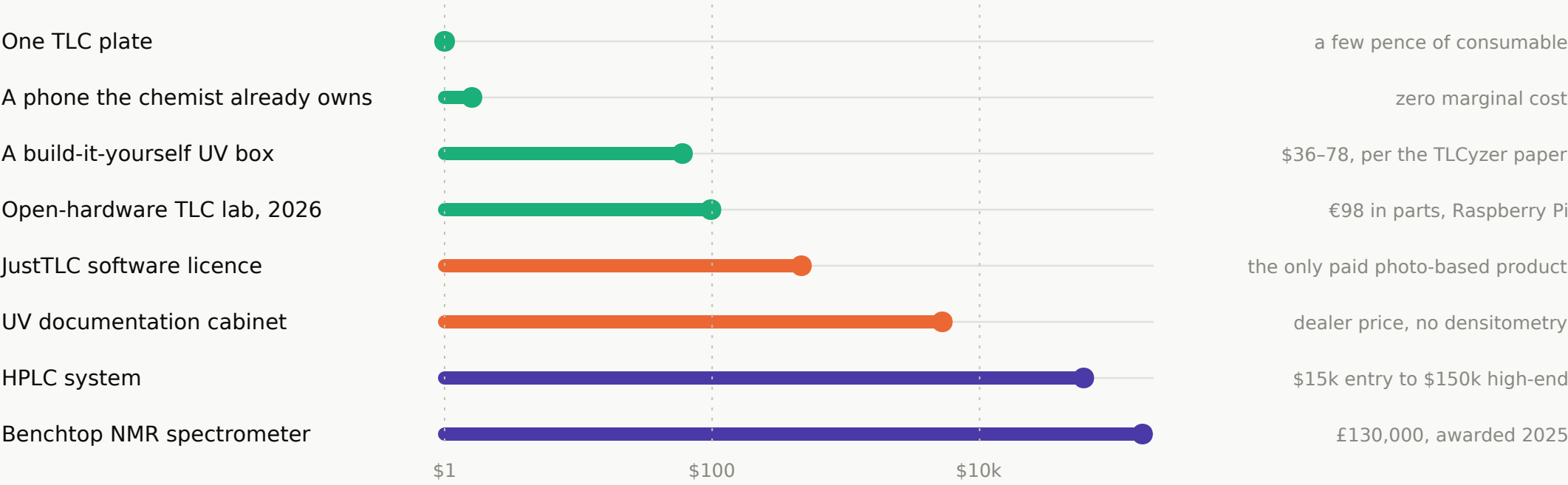
Historically, nobody — and the commercial reasons are worth understanding before we repeat them

Consumables are the business	60–65% of all spending in this market is plates, reagents and standards. A vendor's revenue comes from the plate being used, not from the plate being read.
A quantitative HPTLC system is five instruments	Applicator, chamber, densitometer, imaging box, immersion device, plus server and client licences — before qualification services. No vendor publishes a total price.
An HPLC costs \$15k–\$150k	A lab facing a capital decision buys the instrument that also solves ten other problems. TLC has lived in HPLC's shadow for forty years.
The one paid TLC package is \$465	Less than a day of a chemist's time, and still almost no visible user base.
A whole TLC lab now costs €98 in parts	An academic group published an open-hardware system in 2026. The hardware floor has collapsed; interpretation is what remains scarce.
And TLC itself is a ten-minute job	A flash-chromatography vendor's own marketing: “invest 10 minutes on TLC and save a day of grief.” That is the size of the problem being automated.

THE STRUCTURAL ANSWER

Reading a plate is a small job that happens constantly. Too small to justify a capital instrument, too frequent to send to an analytical queue, too unglamorous to be anybody's flagship. The labs that could pay bought an HPLC instead; the chemists who needed it were handed a free app telling them an Rf they could already estimate by eye. Nobody was wrong — the value simply sat with a buyer nobody was selling to.

THE COST LADDER THIS PROBLEM SITS ON — LOGARITHMIC SCALE



Is there a buyer that didn't exist before?

Robots cannot look at a plate — and about four billion dollars just went into robots that run reactions

WHAT ACTUALLY CLOSES THE LOOP IN A SELF-DRIVING LAB

platform	what it does	instrument that closes the loop	TLC?
A-Lab (Berkeley)	inorganic synthesis; 41 targets in 17 days	powder X-ray diffraction with a neural network for phase identification	no
Chemputer (Glasgow)	programmable synthesis, self-optimising	benchtop NMR in a flow cell, HPLC, Raman, UV-Vis	no
Coscientist (CMU)	a GPT-4 agent planning and running couplings	UV-Vis plate reader and GC-MS	no
Mobile robots (Liverpool)	two robots roaming a normal lab for days	UPLC-MS and benchtop NMR, fused by a decision rule	no
RoboChem-Flex (Amsterdam)	modular low-cost platform, Bayesian optimisation	fluorine NMR, HPLC-MS, or a human running UHPLC by hand	no
Emerald Cloud Lab	remote cloud laboratory, 150+ instruments	UPLC, HPLC, GC-MS, IR, Raman — and a flash system, but no TLC instrument at any level	no

THE IMPORTANT COLUMN IS THE LAST ONE

Not one autonomous chemistry platform in the published record uses TLC as its feedback sensor. The standard methodology review of the field lists HPLC-MS, UV-Vis, photoluminescence and electrochemistry as closed-loop techniques; TLC appears nowhere. That is either the strongest argument against this idea, or the clearest statement of the gap — and which one depends entirely on why TLC is absent.

THE MONEY THAT ARRIVED WHILE NOBODY WAS LOOKING AT TLC

\$2.7bn Isomorphic Labs \$600M Mar 2025, \$2.1B May 2026	\$550m Lila Sciences \$200M seed, then \$350M Series A	\$300m Periodic Labs seed at a \$1.3B valuation, Sep 2025
\$55m Radical AI seed+, Jul 2025 — autonomous materials lab	\$50m+ Chemify Series B, Oct 2025 — commercial Chemputation	\$45m ChemLex Dec 2025 — self-driving lab, Singapore

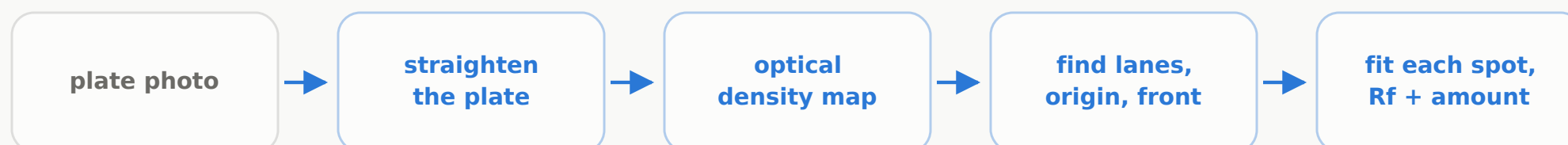
Roughly four billion dollars into this one cohort in eighteen months, and a US executive order in November 2025 explicitly directing the construction of robotic laboratories able to run AI-directed experiments, followed by more than five billion in federal commitments in July 2026. Whatever one thinks of the valuations, demand for machine-readable reaction feedback is being funded on a scale that did not exist when any product in this report was designed.

And the honest counter-argument: those platforms are not waiting for TLC. They already have benchtop NMR at around £130,000, inline infrared and automated HPLC sampling — instruments that give a machine-readable answer without anybody photographing anything. TLC's advantage is that it costs pennies and needs no capital, which matters most to the labs least likely to be building a self-driving one.

So what exactly would we build?

Measure with physics, interpret with language, and never let either cross the line

MEASUREMENT — DETERMINISTIC, NO MODEL, REPEATABLE TO THE PIXEL



THE HANDOVER — a table of numbers, not an image: lane · Rf · optical density · share of lane · signal-to-noise · confidence

INTERPRETATION — A LANGUAGE MODEL, WORKING ONLY FROM THAT TABLE PLUS THE CHEMIST'S CONTEXT

lane names — typed,
or read off the plate

what reaction
this is

what was
expected to form

earlier plates in
the same series

→ named spots · a verdict on the reaction · a recommended next step · a confidence for each

If confidence is low anywhere, the system says so and asks the chemist rather than guessing — and every number stays traceable back to the pixels it came from.

WHY THIS SPLIT, AND NOT ONE MODEL DOING EVERYTHING

It matches where each part is strong

Page 13 measured it: purpose-built vision wins at extraction by wide margins, language models win at explanation and naming.

A wrong measurement is invisible; a wrong name is arguable

If a model quietly reports Rf 0.44 instead of 0.41, nobody can tell. If it says “this is your product” and it is not, a chemist notices immediately and overrules it. Put the fallible component where mistakes are visible.

It is reproducible where reproducibility is required

The measurement path contains no model and no sampling, so the same photograph gives the same numbers every time — a precondition for anything that ends up in a regulated record.

It is what the two published hybrids already do

An atomic-force-microscope agent uses classical thresholding to measure and a language model to plan. An engineering-drawing system uses a detector to find regions and a small fine-tuned model to read them: 93.5 F1, against 61.5 for the model working alone.

WHERE THIS CAN STILL FAIL

Three honest weaknesses. The chemist has to supply the context, which is friction — and friction is what killed every free tool on pages 5 and 6. A language model can produce a confident wrong name, so the interface has to make overruling it trivial. And the whole thing rests on the measurement layer being right: if the geometry fails, the model reasons beautifully about wrong numbers.

Q16 What is it worth, honestly?

Three possible businesses. The certain one is worth little; the valuable one is the least proven.

A

A photo becomes a record

LOW — the market has voted

Photograph the plate, get Rf values, spot intensities and a filed image an electronic notebook can hold.

- Exists already, free, from at least ten groups — pages 5 and 6.
- Value to a chemist: seconds saved, and a record they may not need.
- This business has been tried repeatedly and has almost no users.

B

A reaction becomes a verdict

MEDIUM — real gap, uncertain buyer

Read a plate, or a series of plates, in the context of the reaction and say what happened: consumed, formed, unexplained, done.

- Nobody sells this. The nearest thing is a February 2026 add-on that classifies whole samples.
- Value: the judgement a junior chemist gets wrong and a senior one does in seconds.
- Risk: the senior chemist does it in seconds, and does not want help.

C

A plate becomes a sensor

HIGH IF IT LANDS — furthest from proof

Give an automated platform a cheap, machine-readable answer to “did the reaction work?” without a capital instrument.

- No autonomous platform currently uses TLC. Every one uses NMR, HPLC, MS or XRD.
- Value: the cheapest possible feedback signal for a loop that currently needs a £130k instrument.
- Risk: those platforms already own the expensive instrument, and may never want the cheap one.

The value ordering is the opposite of the certainty ordering. A is certain and worth little; C could be worth a lot and is the least proven thing in this report. B is where the evidence actually points — a real gap, at a level nobody sells, with a buyer we have not yet demonstrated.

AND THE FOUR THINGS WE STILL CANNOT SHOW

Structure on hard photographs

0 of 7 archive plates gave up a solvent front. Our fallback works, but it is a fallback.

Anyone asking for this

Every free tool in this space is unused. We have no evidence yet that our layer is the one people wanted.

Performance outside our own bench

Everything measured in this report was measured on seven plates we chose.

A buyer for the decision layer

The only published system making TLC decisions is a language model reading typed numbers inside somebody else's platform.

How would we know if it worked?

By trying to break it, on plates we did not choose, against a chemist who has not seen our answer

1**Plates we did not choose**

Pull a batch at random from the lab archive, including the badly photographed ones, and run every single one. Report every failure, not a selected set.

2**A chemist as the reference**

A chemist reads each plate blind and records a verdict before seeing ours. Agreement rate is the only number that matters here.

3**Lead with the false alarms**

The Nigeria trial failed on specificity, not sensitivity. Any results page we produce should open with how often we said something was there when it was not.

4**Then hand it to someone who wants it to fail**

Give it to a chemist who did not ask for it, watch where they stop using it, and record that rather than explaining it away.

AND THE THREE THINGS THAT WOULD TELL US TO STOP

A chemist times it and wins

If a chemist reading by eye reaches the same conclusion as our system, as reliably and in less time, the tool has no job on a human bench.

A frontier model closes the gap

If a general vision model starts measuring positions and intensities as well as a purpose-built pipeline, the measurement layer stops being worth owning.

Robots pick a different sensor

If autonomous platforms settle on benchtop NMR or inline analytics, the machine buyer we are aiming at never appears.

WHAT THAT LOOKS LIKE AS A PIECE OF WORK

WEEK 1**pull the archive**

Collect every plate photograph the lab already has, with whatever notebook entry goes with it. No new experiments.

WEEK 2**blind reading**

A chemist reads each one and records a verdict, without seeing any output from us.

WEEK 3**run and compare**

Our system reads the same plates. Publish agreement, disagreement and every outright failure.

WEEK 4**put it in someone's hands**

Someone who did not ask for it uses it on live plates, and we record where they stop.

AND WHAT SUCCESS WOULD ACTUALLY LOOK LIKE, IN NUMBERS

agrees with the chemist

on 9 plates out of 10, on plates nobody curated

false alarms

below 5% — the number the Nigeria trial failed on

effort per plate

one photograph and nothing else typed, or it will not be used

THE ONE-SENTENCE VERSION

Measuring a TLC plate is finished work that at least ten groups already give away for free; naming what is on it, in the context of the reaction it came from, is what nobody sells — and we do not yet know who will pay for it.

SOURCES

Everything in this report, linkable

Checked at source; anything that could not be confirmed is marked in the text instead of cited here

TOOLS THAT READ A PLATE FROM AN IMAGE

· TLC Analyzer (Hess, J. Chem. Educ. 2007, doi 10.1021/ed084p842) — sciencebuddies.org/science-fair-projects/competitions/tlc-analyzer

· qTLC web app (UCL, J. Chem. Educ. 2018, doi 10.1021/acs.jchemed.8b00144) — qtlc.app · discovery.ucl.ac.uk/id/eprint/10061140/

· ImageJ + 3D-printed lightbox for GPHF-Minilab (FHI 360, Analytica 2024) — mdpi.com/2673-4532/5/4/36

· ImageJ / Fiji Gel Analyzer route, community thread — forum.image.sc/t/problem-measuring-relative-amounts-from-scan-of-tlc-plate/31232

· TLCyzer (Hauk et al., Sci. Rep. 2022;12:13433) — nature.com/articles/s41598-022-17527-y · tlczyzer.github.io

· Advion Interchim T.L.C. to Flash & Prep app — flash-chromatography.com/tlc-to-flash-prep-chromatography-app/ · play.google.com/store/apps/details?id=com.AIS.Interchim

· Advion Interchim TLC FlashReader — flash-chromatography.com/tlc-plate-tlc-flashreader/

· Sweday JustTLC and JustQuantify Free — sweday.com

· Chromato (Android + OpenCV) — github.com/ayush-shetty/Chromato · T.L.C Vision — play.google.com/store/apps/details?id=com.tlcvisionn.app

· Sorbfil TLC View flatbed densitometer (Russia) — sorbfil.com · AR2i ChromImage 3X (France)

· TotalLab Phoretix 1D TLC procedure — totallab.com/resources/how-to-analyse-thin-layer-chromatography-tlc-experiments-using-phoretix-1d-image-analysis-software/

· UN-SCAN-IT gel — silksscientific.com/gel/ · GelAnalyzer — gelanalyzer.com · LabImage TLC — kapelanbio.com

· qtlc R package (CRAN) — cran.r-project.org/web/packages/qtlc/ · rTLC (Anal. Chem. 2016, doi 10.1021/acs.analchem.6b04017) · quanTLC (J. Chromatogr. A 2018)

· appaloosa — github.com/marcottelab/appaloosa · pyTLC — github.com/alexandrovteam/pyTLC · TLC_profile — github.com/Pavani-dadi/TLC_profile

· 2LabsToGo-Eco and Office Chromatography (Morlock group) — github.com/OfficeChromatography/2LabsToGo-Eco · Anal. Chem. 2018, doi 10.1021/acs.analchem.8b02866

· Chemotion TLC annotator — tlc.chemotion.ibcs.kit.edu · HPTLC ATLAS — hptlc-association.org/atlas/

INSTRUMENTS

· CAMAG TLC Scanner 4, Visualizer 3, visionCATS, HPTLC PRO — camag.com

· visionCATS 4.2 AI Tools (Feb 2026) — lpp-group.com/news/visioncats-42-introduces-new-ai-tools-for-enhanced-hptlc-data-analysis

· Merck TLC Explorer — sigmaaldrich.com/US/en/campaigns/tlc-explorer-documentation-system

· Biostep / Bionis CD60 — bionis.fr · Syngene ChromaScan — syngene.com/tlc/ · Analytik Jena ChromaDoc-It

· Teledyne ISCO PeakTrak, Biotage Selekt, Büchi Pure TLC-to-flash workflows — all manual Rf entry, per their own manuals

ACCURACY, REPRODUCIBILITY AND METHOD PAPERS

· Sherif et al., dual-detection HPTLC: smartphone imaging vs densitometry, BMC Chem. 2026;20:38 — doi 10.1186/s13065-025-01711-y

· Moaaz et al., smartphone HPTLC vs densitometry, BMC Chem. 2024;18:185 — doi 10.1186/s13065-024-01285-1

· Gad et al., TLC-smartphone for antibiotics, RSC Adv. 2021;11:19196 — doi 10.1039/d1ra01346g

· Comparative study of densitometry and videodensitometry, PRISMA systematic review, 2023 — e-journal.unair.ac.id/JFIKI/article/view/44854

· IAEA-TECDOC-1462, inter-laboratory Rf reproducibility — www-pub.iaea.org/MTCD/Publications/PDF/te_1462_web.pdf

· Humidity and hRf, JPC 2023 — doi 10.1007/s00764-023-00260-1

· Moreira et al., automatic lane segmentation by continuous wavelet transform, Comput. Math. Methods Med. 2013;2013:218415 — doi 10.1155/2013/218415

· Automatic lane and band detection in TLC images, ICIAR 2004 — doi 10.1007/978-3-540-30126-4_20

· Wróbel-Szkolak, Cwener & Komsta, hyperspectral TLC of 70 Polish grasses, Molecules 2023;28:3745 — doi 10.3390/molecules28093745

MACHINE LEARNING ON PLATES, AND ON NUMBERS

· Fichou & Morlock, neural network for planar chromatographic image evaluation, Anal. Chem. 2018 — PMID 29708729

· Stern, Leidig & Wolffe, GAN + CNN botanical identification, J. AOAC Int. 2026;109(2):193 — doi 10.1093/jaoacint/qsae091 · camag.com/article/machine-learning-for-botanical-identification/

· Machine vision quantification of tea saponins from a TLC haemolysis assay, LWT 2024

· Xu et al., robotic TLC platform and 4,944 Rf values, Chem 2022 — cell.com/chem/fulltext/S2451-9294(22)00421-1 · protocol: cell.com/star-protocols/fulltext/S2666-1667(22)00773-0

· Xu et al., TLC to column chromatography, Nat. Commun. 2025;16:832 — nature.com/articles/s41467-025-56136-x

· Liu et al., ML prediction of Rf and tailing propensity, J. Chromatogr. A 2026;1773:466846

· Ruan et al., LLM synthesis platform with an eluent-selection agent, Nat. Commun. 2024;15:10160 — nature.com/articles/s41467-024-54457-x

· Hemmateenejad et al., MIA-TLC reaction monitoring, Anal. Methods 2012;4:933 — doi 10.1039/C2AY25023C

WHAT VISION MODELS CAN AND CANNOT MEASURE

· MeasureBench, arXiv:2510.26865 · live leaderboard flageval-baai.github.io/MeasureBenchPage/

· AECV-Bench, architectural and engineering drawings, arXiv:2601.04819

· CharXiv, NeurIPS 2024 — arxiv.org/abs/2406.18521 · ChartX — arxiv.org/abs/2402.12185

· BLINK, ECCV 2024 — arxiv.org/abs/2404.12390 · GPT-4o vision evaluation, ICLR 2026 — arxiv.org/abs/2507.01955

· Khan et al., hybrid detector + VLM for engineering drawings, RCIM 2025 — arxiv.org/abs/2506.17374

· Mandal et al., LLM agents for atomic force microscopy, Nat. Commun. 2025;16:9104

FIELD USE, MONEY AND POLICY

· Lächele et al., Nigeria field study, J. Pharm. Policy Pract. 2024;17:2432471 — pmc.ncbi.nlm.nih.gov/articles/PMC11632929/

· GPHF-Minilab — gphf.org/en/minilab/ · 13-country Minilab surveillance, Sci. Rep. 2022 — doi 10.1038/s41598-022-17123-0

· A-Lab, Nature 2023 — nature.com/articles/s41586-023-06734-w · Author Correction, Jan 2026 — nature.com/articles/s41586-025-09992-y

· Coscientist, Nature 2023 — nature.com/articles/s41586-023-06792-0 · Mobile robotic chemists, Nature 2024 — nature.com/articles/s41586-024-08173-7

· Seifrid et al., self-driving lab methodology review, Acc. Chem. Res. 2022

· Isomorphic Labs, Lila Sciences, Periodic Labs, Radical AI, Chemify, ChemLex — company announcements 2025–2026

· Executive Order 14363 “Launching the Genesis Mission”, Nov 2025 — whitehouse.gov

· Corkan, Haynes, Kline & Lindsey, robotic TLC instrument, 1991 — doi 10.1007/978-1-4899-0626-7_33